C5.3 Snoop transactions

This section describes each of the snoop transactions and provides information on the recommended behavior where options exist.

The following abbreviations are used for the cache line states:

UC UniqueClean.
UD UniqueDirty.
SC SharedClean.
SD SharedDirty.
I Invalid.

C5.3.1 ReadOnce

Table C5-7 shows all the permitted cache line state changes and the associated PassDirty and IsShared snoop responses for the ReadOnce snoop transaction.

Table C5-7 ReadOnce permitted cache line state changes

Cache line initial state	Cache line end state	Snoop Response	
		PassDirty, CRRESP[2]	IsShared, CRRESP[3]
I	I	0	0
UC	I	0	0
	UC	0	1
	SC	0	1
UD	I	1	0
	UD	0	1
	SC	1	1
	SD	0	1
SC	I	0	0
	SC	0	1
SD	I	1	0
	SC	1	1
	SD	0	1

A ReadOnce snoop transaction is received by a snooped master when the initiating master indicates that it is not going to keep a cached copy of the cache line it is accessing. The ReadOnce snoop transaction enables the snooped master to:

- keep the cache line in a Unique state
- carry out a later store to the cache line without issuing a transaction to obtain permission to store.

If the snooped master has a copy of the cache line, then this specification recommends that data is transferred. If the snooped master has the cache line in a Dirty state, then data must be transferred.

This specification recommends that the cache line is passed as Clean. Although it is permitted to pass the cache line as Dirty, this requires the interconnect to write the cache line back to main memory and the cache line to move to either the SharedClean or Invalid state.

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Note

The IsShared snoop response must be asserted if the snooped master is retaining a copy of the cache line, even if the retained copy is in a Unique state.

C5.3.2 ReadClean, ReadShared, and ReadNotSharedDirty

The ReadClean, ReadShared, and ReadNotSharedDirty snoop transactions have the same requirements, but differ in the behavior that this specification recommends.

Table C5-8 shows all the permitted cache line state changes and the associated PassDirty and IsShared snoop responses for these snoop transactions.

Table C5-8 ReadClean, ReadShared, and ReadNotSharedDirty permitted cache line state changes

Initial state	End state	Snoop Response	
		PassDirty, CRRESP[2]	IsShared, CRRESP[3]
I	I	0	0
UC	I	0	0
	SC	0	1
UD	I	1	0
	SC	1	1
	SD	0	1
SC	I	0	0
	SC	0	1
SD	I	1	0
	SC	1	1
	SD	0	1

If the cached line being snooped is part of an exclusive sequence then the cache line must remain valid in the snooped master.

If data is available, this specification recommends that the data is transferred.

ReadClean

For a ReadClean snoop transaction, if the responsibility for writing the cache line back to main memory is being passed to the interconnect, as indicated by the PassDirty snoop response being asserted, the cache line is written back to main memory immediately. This specification recommends that the cache line remains Dirty in the snooped cache.

ReadShared

For a ReadShared snoop transaction, if the responsibility for writing the cache line back to main memory is being passed to the initiating master, then it is accepted by the master. The decision to pass responsibility for writing the cache line that is Dirty back to main memory depends on which master accesses the cache line next:

- If the snooped master is likely to be the next master to store to the cache line, then this specification recommends that the cache line remains Dirty in the snooped cache but is passed as Clean to the initiating master.

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- If the initiating master is likely to be the next master to store to the cache line, then this specification recommends that the cache line is passed to the initiating master as Dirty. In this case:
  - if it is likely that the initiating master carries out a store before the snooped master next loads from the cache line, then this specification recommends that the snooped master does not retain a cached copy
  - if it is likely that the snooped cache loads from the cache line before the initiating master performs a store, then this specification recommends that the snooped master does retain a copy of the cache line.
- If it is not known whether the initiating master or the snooped master is the next to store to the cache line, then this specification recommends that the cache line is held as Dirty in the cache that is least likely to evict the cache line. Typically, this would be the initiating master, because this is the master that has most recently accessed the cache line.
- If the snooped master does not support all five cache states, then less options are available.

If information on the access patterns for a cache line is not available, then this specification recommends that the cache line is passed as Dirty to the initiating master and moves to the SharedClean state in the snooped cache.

ReadNotSharedDirty

If responsibility for updating main memory is passed to the initiating master, it is only accepted if the cache line moves to the Invalid state in the snooped cache. The decision to pass responsibility for writing the cache line back to main memory depends on which master accesses the cache line next:

- if the snooped master is likely to be the next master to store to the cache line, then this specification recommends that the cache line remains Dirty in the snooped cache but is passed as Clean to the initiating master
- if the initiating master is likely to be the next master to store to the cache line, then this specification recommends that the cache line is passed to the initiating master as Dirty and the cache line is removed from the snooped cache.

If it is not known which master accesses the cache line next, then no recommendations are provided by this specification.

C5.3.3 ReadUnique

Table C5-9 shows all the permitted cache line state changes and the associated PassDirty and IsShared snoop responses for the ReadUnique snoop transaction.

Table C5-9 ReadUnique permitted cache line state changes

Initial state	End state	Snoop response	
		PassDirty, CRRESP[2]	IsShared, CRRESP[3]
I	I	0	0
UC	I	0	0
UD	I	1	0
SC	I	0	0
SD	I	1	0

For a ReadUnique transaction, if the snooped cache holds a copy of the cache line in a Dirty state, then the data must be transferred.

This specification recommends that data is also transferred if the cache line is in a Clean state.

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C5.3.4 CleanInvalid

Table C5-10 shows all the permitted cache line state changes and the associated PassDirty and IsShared snoop responses for the CleanInvalid snoop transaction.

Table C5-10 CleanInvalid permitted cache line state changes

Initial state	End state	Snoop response	
		PassDirty, CRRESP[2]	IsShared, CRRESP[3]
I	I	0	0
UC	I	0	0
UD	I	1	0
SC	I	0	0
SD	I	1	0

For a CleanInvalid transaction, if the snooped cache holds a copy of the cache line in a Dirty state, then the data must be transferred.

This specification recommends that data is not transferred if the cache line is in a Clean state.

C5.3.5 MakeInvalid

Table C5-11 shows all the permitted cache line state changes and the associated PassDirty and IsShared snoop responses for the MakeInvalid snoop transaction.

Table C5-11 MakeInvalid permitted cache line state changes

Initial state	End state	Snoop response	
		PassDirty, CRRESP[2]	IsShared, CRRESP[3]
I	I	0	0
UC	I	0	0
UD	I	0	0
		1	0
SC	I	0	0
SD	I	0	0
		1	0

For a MakeInvalid transaction, this specification recommends that the data is not transferred.

Note

If data is not transferred, as indicated by the DataTransfer snoop response, **CRRESP[0]** being deasserted, then the PassDirty snoop response bit must also be deasserted.

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C5.3.6 CleanShared

Table C5-12 shows all the permitted cache line state changes and the associated PassDirty and IsShared snoop responses for the CleanShared snoop transaction.

Table C5-12 CleanShared permitted cache line state changes

Initial state	End state	Snoop response	
		PassDirty, CRRESP[2]	IsShared, CRRESP[3]
I	I	0	0
UC	I	0	0
	UC	0	1
	SC	0	1
UD	I	1	0
	SC	1	1
SC	I	0	0
	SC	0	1
SD	I	1	0
	SC	1	1

For a CleanShared transaction, if the snooped cache holds a copy of the cache line in a Dirty state, then the data must be transferred.

This specification recommends that the data is not transferred if the cache line is in a Clean state.

Note

The IsShared snoop response must be asserted if the snooped master is retaining a copy of the cache line, even if the retained copy is in a Unique state.

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Chapter C6
Interconnect Requirements

This chapter describes the interconnect requirements for ACE. It contains the following sections:

- *About the interconnect requirements* on page C6-226
- *Sequencing transactions* on page C6-227
- *Issuing snoop transactions* on page C6-229
- *Transaction responses from the interconnect* on page C6-232
- *Interactions with main memory* on page C6-234
- *Other requirements* on page C6-237
- *Interoperability considerations* on page C6-239.

This chapter does not describe the interconnect requirements for barriers or DVM operations. See Chapter C8 *Barrier Transactions* and Chapter C12 *Distributed Virtual Memory Transactions* for these requirements.

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